

Ali Bahr

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EDUCATION

Cairo University - Faculty of Engineering Oct. 2021 – July 2026
B.Sc. Computer Engineering **GPA: 3.7 / 4.0** Giza, Egypt
Ranked **401st nationwide (top 0.08%)** in Egypt's national secondary-school certificate exam. Aug. 2021

EXPERIENCE

Raisa Energy LLC July 2025 – Oct. 2025
Software Engineering Intern Cairo, Egypt
AI Integration Project

- Automated support-ticket routing by building an LLM classification system that assigns tickets by urgency and responsible team, reducing manual processing costs by **80%**.
- Built a RAG pipeline that injected semantically relevant historical tickets as context, pushing classification accuracy from **70% to 97%**.
- Implemented Azure AI Search as the vector store and built an Azure Function to continuously synchronize the index.
- Tech Stack: OpenAI, Azure AI Search, Azure Functions, RAG, Embeddings, Power Automate*

Backend Project

- Introduced a cursor-based pagination system across all API resources, cutting page load time **~30x** (from ~9s to under 300ms).
- Migrated filtering, sorting, and search features from the frontend to the backend with a generic, reusable query-building architecture applied consistently across all resources.
- Tech Stack: ASP.NET Core, EF Core, SQL Server*

TCCD Cairo University Aug. 2024 – Aug. 2025
Backend Developer, IT Committee Giza, Egypt

- Developed an event registration feature serving **10,000+ attendees**.
- Implemented secure QR code ticket validation for event check-ins, streamlining access control for committee members with role-based authentication.
- Tech Stack: C#, .NET, PostgreSQL, Clean Architecture*

Scale AI June 2023 – Aug. 2023
Software Engineering Contributor Remote

- Authored reference solutions for 10+ competitive programming problems used as training data for code-generation models.
- Created 80+ test cases to validate solution robustness and correctness.

SKILLS

Languages: C++/C, C#, SQL, JavaScript/TypeScript, Python.
Backend: ASP.NET Core, EF Core, Clean Architecture, Node.js/Express, REST APIs.
Databases: SQL Server, PostgreSQL, MongoDB, Redis.
Tools: Git, Linux, Azure, AWS (S3).

ACTIVITIES AND COMPETITIONS

The Egyptian Collegiate Programming Championship (ECPC)

- Solved **1,500+** algorithmic problems across online judges.
- Competed in **ECPC** qualifications 2023, 2024, and 2025.

PROJECTS

Telegrammy | *JavaScript, Node.js, Express.js, MongoDB, Socket.io, WebRTC, Redis, AWS S3*

- Built the real-time messaging server for a team-built Telegram clone on an event-driven Socket.io architecture, exposing all messaging features, with Redis caching.
- Implemented WebRTC voice and group calls end-to-end, covering signaling (offer/answer/ICE exchange) and call state management.
- Developed a profile & privacy management module and a stories module with media uploads to AWS S3.
- Achieved **80%+ test coverage**, ensuring high performance, maintainability, and seamless system integration.

ExploreX | *Java, Spring Boot, MongoDB*

- Designed an advanced search engine capable of crawling URLs, indexing documents, ranking the results, and performing complex queries and phrase searches.
- Implemented a ranking algorithm to optimize search relevance, significantly improving query accuracy, and leveraged parallel processing techniques to reduce average response time **from 9 seconds to less than 1 second**.

VecDB | *Python, Algorithms, Vector Databases*

- Architected a high-performance vector database scaling up to **20M vectors**, achieving sub-second retrieval latency and **100% recall** with highly optimized resource utilization.
- Evaluated performance tradeoffs across HNSW, standard IVF, and IVF-PQ, ultimately implementing a custom 2-Level IVF structure to optimize the balance between search latency and retrieval accuracy.

ScheduVerse | *C, Operating Systems*

- Built a multi-algorithm OS scheduler that manages multiple processes with varying time and memory requirements.
- Enhanced process synchronization with semaphores and advanced IPC techniques.

Signfy | *Python, MediaPipe, TCN, Gemini API, TTS, Poetry* | **Graduation Project**

- Led a 4-engineer team building a real-time, two-way Arabic Sign Language translation system (speech-to-sign and sign-to-speech), owning the sign-to-speech pipeline end-to-end.
- Built landmark extraction and preprocessing (MediaPipe, normalization) and a motion-gated segmentation module that isolates individual signs from continuous video.
- Trained RF and TCN classifiers over a **502-sign vocabulary** and engineered a Gemini prompt converting classifier outputs into grammatically correct, diacritized Arabic sentences, voiced via a pre-trained TTS model (Meta MMS).